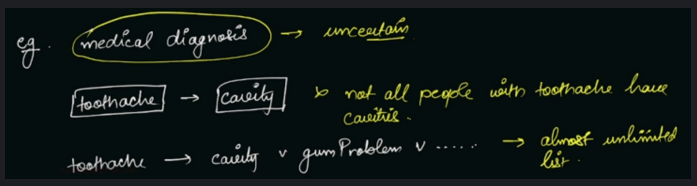
॥श्रीगणेशाय नमः॥

Uncertainty

- Agents in the real world need to handle uncertainty (non-deterministic environments).
- A rational decision requires:
  - goal
  - its likelihood



- If we want a correct rule, then it should be logically exhaustive.
- It fails due to 3 main reasons:
  - Laziness: Too much work.
  - Theoretical ignorance: Incomplete theory of the domain.
  - Practical ignorance: All possible tests can't be run on a patient.

So we need to deal with the degree of belief.

Probability

- Sample space: Set of all possible events.
- Mutually exclusive events: Events that can't occur simultaneously.
- Exhaustive events: Their union covers the sample space.
- Unconditional probability: Probability prior to finding evidence.
- Conditional probability: (Posterior) Belief after finding evidence.

$$P(\text{cavity} \mid \text{toothache}) = 0.7$$

whenever toothache is true, conclude that cavity is true with prob. 0.7 ✗
whenever toothache is true and we have no further info, conclude cavity is true with prob. 0.7.

$$P(A \mid B) = \frac{P(A, B)}{P(B)}$$

$$P(A, B) = P(A \mid B) \cdot P(B) \quad \text{product rule}$$

- Random variable: That represents the outcome of a random process.
  - Discrete
  - Continuous

- Marginal probability: Probability of a single event without considering other events (meaning with all possible combinations of other events).

$$P(X = x) = \sum_y P(X = x, Y = y)$$

$$= \int_{-\infty}^{\infty} f_{x,y}(x, y)\, dy$$

eg.

	toothache		¬ toothache	
	catch	¬ catch	catch	¬ catch
cavity	0.108	0.012	0.072	0.008
¬ cavity	0.016	0.064	0.144	0.576

P(cavity) = ?
P(cavity / toothache)

→ $P(\text{cavity}) = 0.108 + 0.012 + 0.072 + 0.008 = 0.2$

$$P(\text{cavity} \mid \text{toothache}) = \frac{P(\text{cavity} \cap \text{toothache})}{P(\text{toothache})} = \frac{0.108 + 0.012}{0.2} = \frac{0.12}{0.2} = 0.6$$

$$P(\neg \text{cavity} \mid \text{toothache}) = \frac{0.016 + 0.064}{0.108 + 0.012 + 0.016 + 0.064} = 0.4 = 1 - 0.6$$

- Bayes theorem:

$$P(\text{cause} \mid \text{effect}) = \frac{P(\text{effect} \mid \text{cause}) \cdot P(\text{cause})}{P(\text{effect})}$$

- Naive Bayes models:

$$P(\text{cause} \mid \text{effect}_1, \text{effect}_2, \ldots, \text{effect}_n) = P(\text{cause}) \cdot \prod_i P(\text{effect}_i \mid \text{cause})$$

Naive Bayes assumption: all features are conditionally independent.

$$P(c \mid e_1 \cdots e_n) = \frac{P(c) \cdot \prod_i^n P(e_i \mid c)}{P(e_1 \cdots e_n)}$$

(posterior

Bayesian networks
(Belief networks)

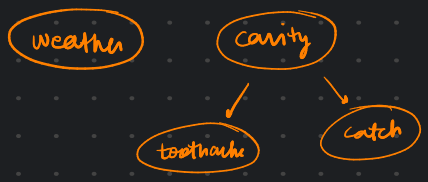
- Each node denotes a random variable.
- Arrows that connect these nodes:



these networks are DAGs.

r.v. X has direct influence on r.v. Y.

X: cause Y: effect



cavity: there is a cavity
catch: probe catches cavity

→ weather is independent of the other variables.
→ Toothache and catch are conditionally independent given cavity.

- Each node x_i is associated with a probability information that quantifies the effect of parents on the nodes using a finite # of parameters.

Burglars and earthquakes

BNs: Qualitative Structure

- Graphical structure of BN reflects conditional independence among variables
- Each variable *X* is a node in the DAG
- Edges denote *direct probabilistic influence*
  - usually interpreted *causally*
  - parents of *X* are denoted *Par(X)*
- ***Local semantics:* X is conditionally independent of all nondescendents given its parents**
  - Graphical test exists for more general independence
  - "Markov Blanket"

Burglars and Earthquakes

- You are at a "Done with the AI class" party.
- Neighbor John calls to say your home alarm has gone off (but neighbor Mary doesn't).
- Sometimes your alarm is set off by minor earthquakes.
- Question: Is your home being burglarized?
- Variables: Burglary, Earthquake, Alarm, JohnCalls, MaryCalls
- Network topology reflects "causal" knowledge:
  - A burglar can set the alarm off
  - An earthquake can set the alarm off
  - The alarm can cause Mary to call
  - The alarm can cause John to call

Example

- Pearl lives in Los Angeles. It is a high-crime area. Pearl installed a burglar alarm. He asked his neighbors John & Mary to call him if they hear the alarm. This way he can come home if there is a burglary. Los Angeles is also earth-quake prone. Alarm goes off when there is an earth-quake.

Burglary => Alarm
Earth-Quake => Alarm
Alarm => John-calls
Alarm => Mary-calls

If there is a burglary, will Mary call?
Check KB & E |= M
If Mary didn't call, is it possible that Burglary occurred?
Check KB & ~M *doesn't entail* ~B

How do we handle Real Pearl?



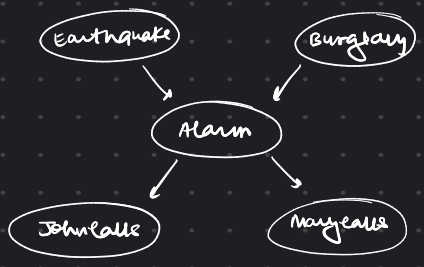
Potato in the tail-pipe

Omniscient & Eager way:
- Model everything!
- E.g. Model exactly the conditions under which John will call
  - He shouldn't be listening to loud music, he hasn't gone on an errand, he didn't recently have a tiff with Pearl etc etc.

A & c1 & c2 & c3 &..cn => J
(also the exceptions may have interactions
c1&c5 => ~c9)

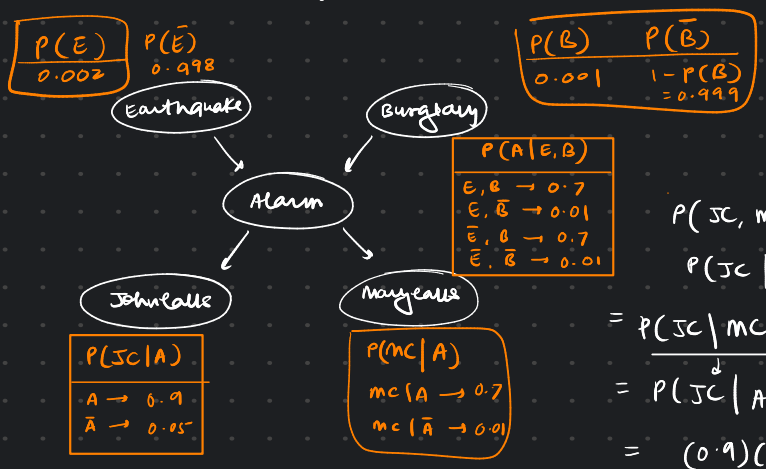
Qualification and Ramification problems make this an infeasible enterprise

- Ignorant (non-omniscient) and Lazy (non-omnipotent) way:
  - Model the likelihood
  - In 85% of the worlds where there was an alarm, John will actually call
  - How do we do this?
    - Non-monotonic logics
    - "certainty factors"
    - "fuzzy logic"
    - "probability" theory?



5 r.v.s here, so 5 conditional prob. tables are reqd.

Conditional probability table (CPT)



$$P(JC, MC, A, E, B) =$$
$$P(JC \mid MC, A, E, B) \cdot P(MC, A, E, B)$$
$$= P(JC \mid MC, A, E, B) \cdot P(MC \mid A, E, B) \cdot P(A \mid E, B) \cdot P(E \mid B) \cdot P(B)$$
$$= P(JC \mid A) \cdot P(MC \mid A) \cdot P(A \mid E, B) \cdot P(E) \cdot P(B)$$
$$= (0.9)(0.7)(0.7)(0.002)(0.001)$$

- Only 10 distinct parameters are required to have a full joint probability distribution than having 31.

$(2^5 - 1)$

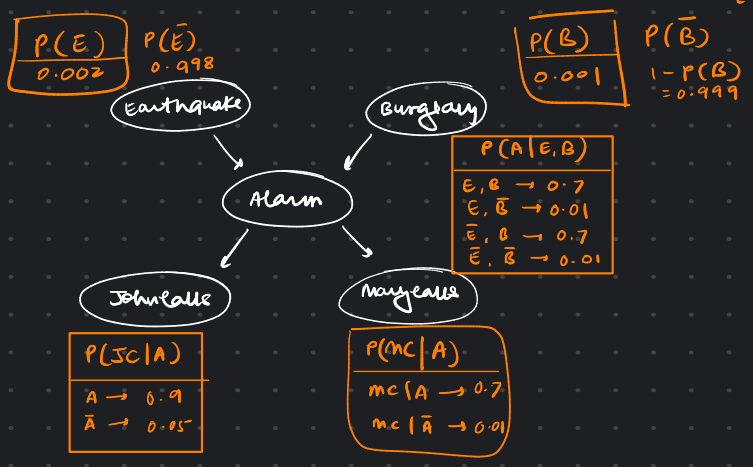
Theorem of factorization

- The full joint distribution of a Bayes net is $$P(X_1, X2, \ldots, X_n) = \prod_{i=1}^{n} P(X_i | Par(X_i))$$

$Par(x_i) \rightarrow$ parents(x_i)

Q. Find $P(J, M, A, \neg B, \neg E)$.

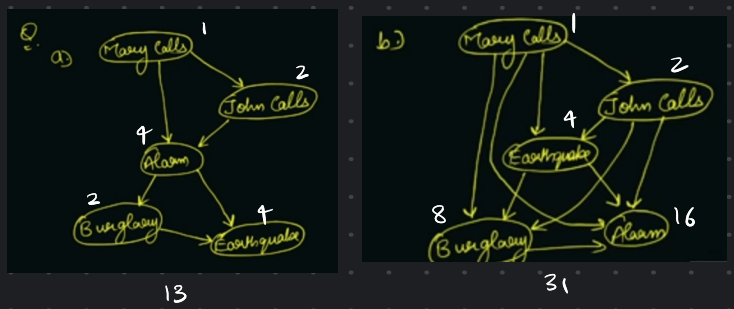
complement | parents can be calculated



$\rightarrow P(J, M, A, \bar{B}, \bar{E}) = P(J|A) \cdot P(M|A) \cdot P(A|\bar{B}, \bar{E}) \cdot P(\bar{B}) \cdot P(\bar{E})$

$= (0.9)(0.7)(0.01)(0.999)(0.998) = 0.00628$

Q.

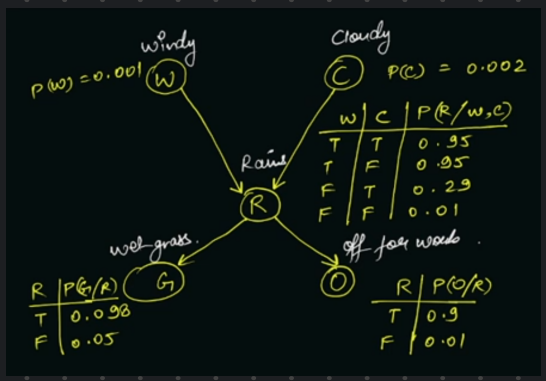


13 31

Given these Bayes nets, how many distinct probs are reqd to calculate the full joint dist^n?

For any node $\Rightarrow 2^{\#\text{ of parents}}$

Q.



Prob. of having wet grass.
$P(G) = ?$

$\rightarrow P(G) = P(G|R) \cdot P(R) + P(G|\bar{R}) \cdot P(\bar{R})$

$P(R) = P(R|W,C) \cdot P(W,C) + P(R|\bar{W},\bar{C}) \cdot P(\bar{W},\bar{C}) + P(R|\bar{W},C) \cdot P(\bar{W},C) + P(R|W,\bar{C}) \cdot P(W,\bar{C})$

$= (0.95)(0.001)(0.002) + (0.01)(0.999)(0.998) + (0.95)(0.999)(0.002) + (0.29)(0.001)(0.998)$

$= 0.0122$

$P(\bar{R}) = 0.9878$

$P(G) = (0.098)(0.0122) + (0.05)(0.9878) \approx 0.0506$

Conditional independence

directional

d-Separation

Are {X} and {Y} dependent given {Z} ?

⊥

- An undirected path between 2 nodes is blocked (cut-off): information cannot flow from X -> Y, thus they are d-separated or conditionally independent.
- 2 nodes are d-separated, if all undirected paths are blocked.

Steps

1. Draw an ancestral graph.

{X, Y, Z}

graph consisting the mentioned variables (in X, Y, Z) and all their ancestors.



Eg.

Ancestral graph of set {A}:



Steps:
① draw ancestral graph
② moralize (marry parents)
③ disorient (undirected edges)
④ delete (given)

2. Moralize: moralize the ancestral graph by marrying the parents.

connect all possible pairs of parents with undirected edges.





moralize:
- go to each node.
- If a node has > 1 parent, connect them.
- make all pairwise connections.
- do this for all nodes.

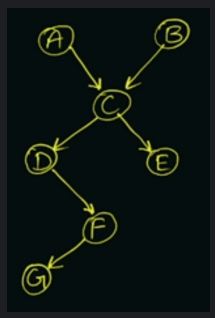
3. Disorient: Replace all directed edges by undirected ones.



4. Deletion: Delete the givens (all nodes in Z) and their edges.

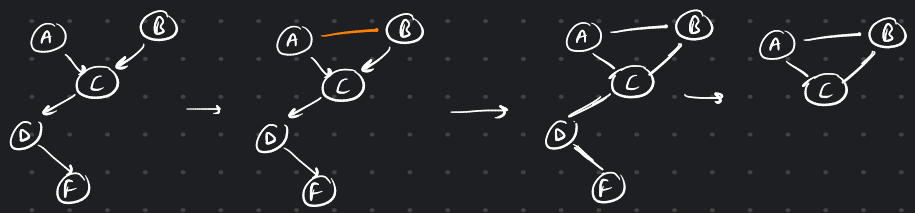
5. Conclusion:
If X and Y are disconnected, they are d-separated, thus conditionally independent given Z.
If they are connected, we can't guarantee conditional independence.

Q.



1. Are (A) & (B) conditionally independent, given (D & F) ? P(A/B,D,F) = P(A/D,F)
2. Are A & B marginally independent ?
3. X = {A}, Y = {B}, given Z = {C}.

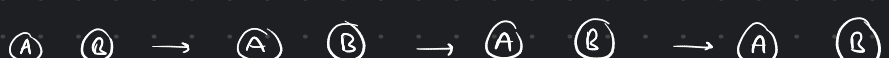
→ 1. X = {A}
Y = {B}
Z = {D, F}



A and B are d-connected.

dependent

2. X = {A}, Y = {B}, Z = ϕ



d-separated

independent

3.



dep.

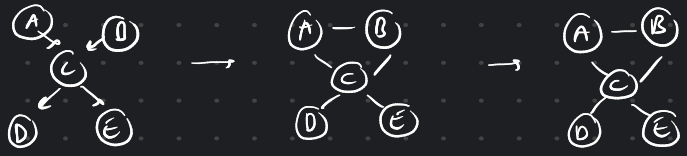
4. D & E, given C ?
5. D & E → marginally independent ?

4.



indep.

5.

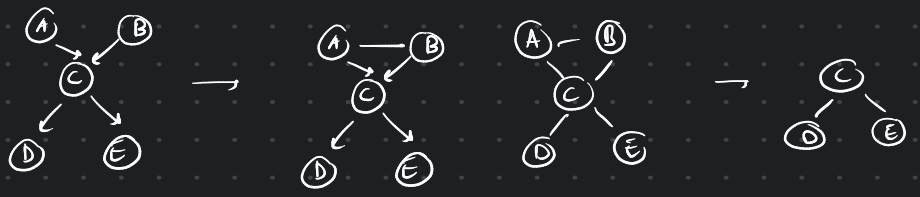


dep.

dependent even if there's indirect link.

6. D & E, given A & B ?
7. If $P(D/EGC) = P(D/C)$?

6.

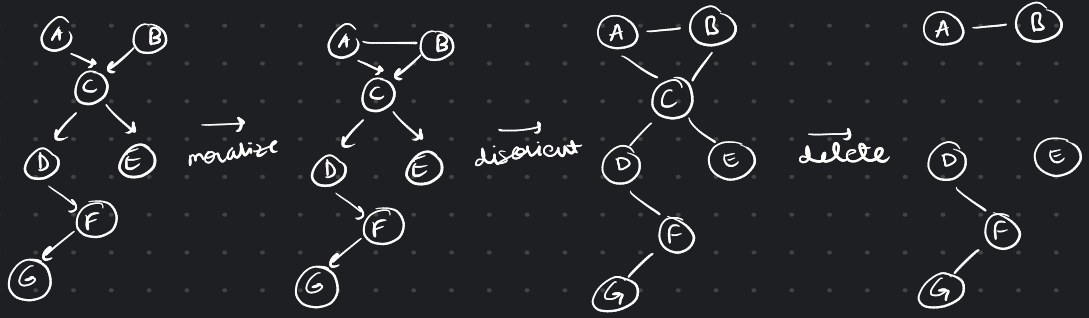


dep.

In case of X and Y having multiple variables, check all pairs.
X_i must be d-separated from all Y_j $\forall i, j$.

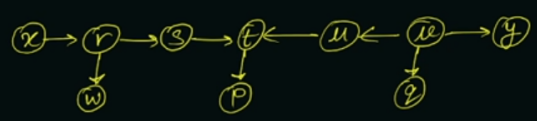
7. If D is ind. of both E & G given C ?

$X = \{D\}$ $Y = \{E, G\}$ $Z = \{C\}$



D and E should be disconnected. ✓
↑
D and G must be disconnected ✗
No
=
X is not ind. of Y given C.

(2)

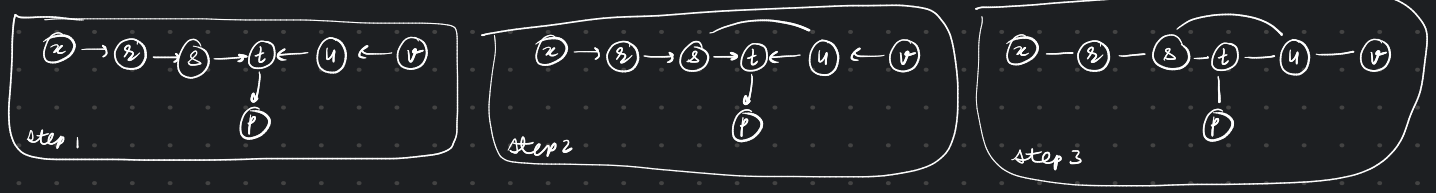


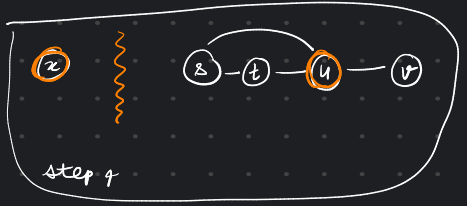
$Z = \{r, p\}$ → given.

a) $X = \{x\}$, $Y = \{u\}$
b) $X = \{s\}$, $Y = \{y\}$

} Independent, given Z ?

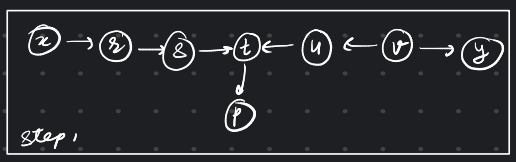
a)

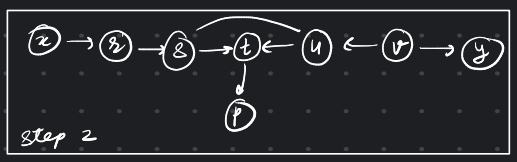


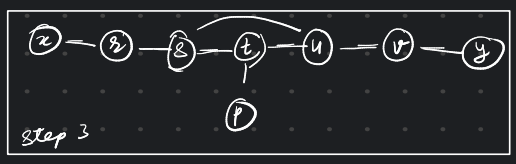


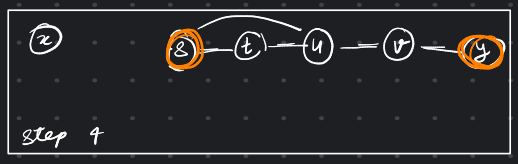
Independent

b)









not d separated.
dependent

Inference in Bayes nets

- Given a BN, we are interested in 2 types of probabilities:
  1. Marginal 2. Conditional/ Posterior

Inference → exact : directly computing the prob ; enumeration (direct computation), variable elimination (more efficient)
Inference → approximate : some approximations are used

Exact inference

$P(J, M, A, E, B)$ = known

To compute $P(M, A, E, B)$ → take marginal prob. over J.

$$P(M, A, E, B) = \sum_J P(J, M, A, E, B)$$

Enumeration:

$P(B \mid J, M)$ = Prob. of burglary given mary and John called.

$$\Rightarrow P(B \mid J, M) = \frac{P(B, J, M)}{P(J, M)} \quad \text{——— (1)}$$

could replace with → $P(J|A) \cdot P(M|A) \cdot P(A|B,E) \cdot P(E) \cdot P(B)$ by factorization

$$\text{and } P(J, M) = \sum_{E,B,A} P(E, B, A, J, M)$$

$$\text{and } P(B, J, M) = \sum_{E,A} P(E, B, A, J, M)$$

$P(\{X\} \mid \{Y\}) = \frac{P\{X, Y\}}{P\{Y\}}$] find both separately like marginal probability.

Variable elimination:

$$P(B \mid J, M) = \frac{P(B, J, M)}{P(J, M)} = \alpha \cdot P(B, J, M)$$

$$\alpha = \frac{1}{P(J, M)} = \frac{1}{P(B, J, M) + P(\neg B, J, M)}$$

- In variable elimination, we will be taking the summation inside.

$$\sum_{M,A,B,E} = \sum_M \sum_A \sum_B \sum_E$$

e.g. $\sum_{M,A,B,E} P(B) \cdot P(E) \cdot P(A|BE) \cdot P(J|A) \cdot P(M|A)$

(we pick an order M, A, E, B

Take Σ's inside

$$\Rightarrow \sum_{A,B,E} P(B) \cdot P(E) \cdot P(A|BE) \cdot P(J|A) \sum_M P(M|A)$$

$$\Rightarrow \sum_{B,E} P(B) \; P(E) \sum_A P(A|BE) \cdot P(J|A) \sum_M P(M|A)$$

start variable elimination from the right.

$$\Rightarrow \sum_B P(B) \cdot \sum_E P(E) \cdot \sum_A P(A|BE) \cdot P(J|A) \cdot \underbrace{\sum_M P(M|A)}_{f(A)}$$

$$\Rightarrow \sum_B P(B) \sum_E P(E) \underbrace{\sum_A P(A|BE)\, P(J|A)\, f_1(A)}_{\hookrightarrow f(B,E,J)}$$

$$\Rightarrow \sum_B P(B) \sum_E P(E)\, f_2(B,E,J)$$

$$\Rightarrow \sum_B P(B) \cdot f_3(B,J)$$

$$\Rightarrow f_4(J)$$

Operations on factors

→ 2^n values

- Factor: A function from some random variable to a number.

E.g. $f(x_1, x_2, x_3)$ ~ 2^3 E.g. $f(x_1 \mid x_2, x_3)$

A factor f on random variables x_1, x_2, x_3.

1. Restricting a factor

$f(x_1, x_2, \ldots, x_n) \rightarrow$ restricting f to $x_1 = v_1$

$\Rightarrow f(v_1, x_2, \ldots, x_n)$ over $x_2, \ldots, x_n$

function of x, y, z → $x=1$ → func^n of y and z

$f(x, y, z) \longrightarrow f(0, y, z)$

$f_1(X, Y, Z)$

X	Y	Z	f_1 val
T	T	T	0.1
T	T	F	0.9
T	F	T	0.2
T	F	F	0.8
F	T	T	0.4
F	T	F	0.6
F	F	T	0.3
F	F	F	0.7

Find, (uppercase = R.V., lowercase, T)

a) $f_2(Y, Z) = f_1(x, Y, Z)$

b) $f_3(Y) = f_2(Y, \neg z)$ → F

c) $f_4() = f_3(\neg y)$

a) Restrict x to T

$f_2(Y, Z)$:

Y	Z	f_2 value
T	T	0.1
T	F	0.9
F	T	0.2
F	F	0.8

b) $f_3(Y)$:

Y	val.
T	0.9
F	0.8

c) $f_4()$:

val.
0.8

2. Sum

- summing out a variable

$f(x_1, x_2, \ldots, x_n)$

$$\sum_{x_1} f(x_1, \cdots, x_n) = f_2(x_2, x_3, \ldots, x_n)$$

$$= f_1(x_1 = T, x_2, \ldots, x_n) + f_1(x_1 = F, x_2, \ldots, x_n)$$

$f_1(X, Y, Z)$

X	Y	Z	f_1 val
T	T	T	0.1
T	T	F	0.9
T	F	T	0.2
T	F	F	0.8
F	T	T	0.4
F	T	F	0.6
F	F	T	0.3
F	F	F	0.7

$f_1(X, Y, Z)$ given, find $f_2(X, Z) = \sum_Y f_1(X, Y, Z)$?

→ $f_2(X, Z) = f_1(X, Y = T, Z) + f_1(X, Y = F, Z)$

X	Z	val
T	T	0.1 + 0.2 = 0.3
T	F	0.9 + 0.8 = 1.7
F	T	0.4 + 0.3 = 0.7
F	F	0.6 + 0.7 = 1.3

3. Multiplying 2 factors

$f_1 \times f_2 = f_1 * f_2$

	f_1	×	f_2	=	$f_1 * f_2$
case 1:	X, Y		X, Y		X, Y
case 2:	X, Y		Y, Z		X, Y, Z
case 3:	X, Y		A, B		X, Y, A, B

$f_1(x, y) \rightarrow$

X	Y	val
T	T	0.1
T	F	0.9
F	T	0.2
F	F	0.8

$f_2(Y, Z) \rightarrow$

Y	Z	val
T	T	0.3
T	F	0.7
F	T	0.6
F	F	0.4

$f_3(x, y, z) = f_1 \times f_2$

f_3 :

x	y	z	value
T	T	T	
T	T	F	$0.1 \times 0.7 = 0.07$
T	F	T	
T	F	F	
F	T	T	
F	T	F	
F	F	T	
F	F	F	

eg. $f_1(x, y) \rightarrow$

X	Y	val
T	T	0.1
T	F	0.2
F	T	0.3
F	F	0.4

$f_2(z) \rightarrow$

Z	val
T	0.1
F	0.9

find $f_3(x, y, z) = f_1 \times f_2$

Q. $f(x, y) \rightarrow$

X	Y	val
T	T	0.3
T	F	0.7
F	T	0.9
F	F	0.1

$g(Y, Z) \rightarrow$

Y	Z	val
T	T	0.2
T	F	0.8
F	T	0.6
F	F	0.4

a) $h(x, y, z) = f * g$

b) $h_2(Y, Z) = \sum_x h(x, y, z)$.

$h(x, y, z)$: $f * g$

X	Y	Z	val
F	F	F	0.04
F	F	T	0.06
F	T	F	0.72
F	T	T	0.18
T	F	F	0.28
T	F	T	0.42
T	T	F	0.24
T	T	T	0.06

$h_2(Y, Z)$:

Y	Z	val.
T	T	0.24
T	F	0.96
F	T	0.48
F	F	0.32

Note :- $\sum_x f(X,Y) \times g(Y,Z) = g(Y,Z) \cdot \sum_x f(X,Y)$

do not multiply since g does not involve X.

$\hookrightarrow f_1(Y)$

∴ first calculate $\sum_x f(X,Y)$

$= g(Y,Z) \times f_1(Y)$

4. Normalize a factor

$f_1 \rightarrow$

X	val
T	0.2
F	0.6

$f_1(X=T) + f_1(X=F) \neq 1$

f_2 :

X	val
T	0.25
F	0.75

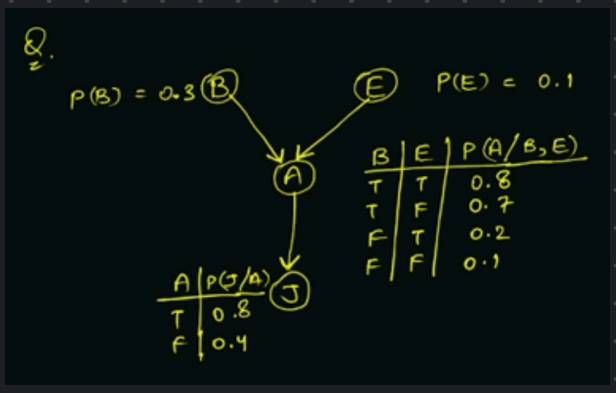
} normalized values.

$$f_2(X=T) = \frac{f_1(X=T)}{f_1(X=T) + f_1(X=F)} = \frac{0.2}{0.2+0.6} = 0.25$$

Algorithm for variable elimination

- The idea is to find any posterior probability, involving the random variables given in the BN.
- The steps are:

1. Construct factors for each probability distribution given in the expression.
2. Restrict the observed variables to their observed values.
3. Eliminate each hidden variable. Apply multiply and sum, starting from the right side.
4. Multiply the remaining factors.
5. Normalize the resulting factor.



Find $P(B / \neg A)$

query variable → B
givens / evidence variables → {A}
hidden variables → {E, J}
(not query nor givens)

→ $P(B|\sim A) = \frac{P(B, \sim A)}{P(\sim A)} = \alpha \cdot P(B, \sim A)$ $\quad \alpha = \frac{1}{P(B, \sim A) + P(\sim B, \sim A)}$

↳ The problem is now to find $P(B, \sim A)$ i.e. $P(B|A=F)$

$$P(B, A=F) = \sum_{E,J} P(B, A=F, E, J)$$

factorization

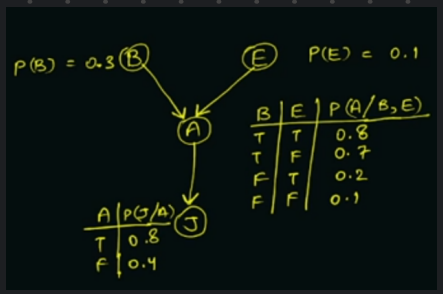


$$= \sum_{E,J} P(B) \cdot P(E) \cdot P(A=F|E,B) \cdot P(J|A=F)$$

$$= \underbrace{P(B)}_{f_1(B)} \sum_{E} \underbrace{P(E)}_{f_2(E)} \cdot \underbrace{P(a=F|E,B)}_{f_3(E,B)\ f_3(A,B,E)} \sum_{J} \underbrace{P(J|a=F)}_{f_4(J)}$$

step 1: Defining factors.

4 initial factors here



$f_1(B) = P(B)$:

B	f1 val
T	0.3
F	0.7

$f_2(E) = P(E)$

E	val
T	0.1
F	0.9

$f_3(A,B,E)$: $P(A|B,E)$

A	B	E	value
T	T	T	0.8
T	T	F	0.7
T	F	T	0.2
T	F	F	0.1
F	T	T	1-0.8=0.2
F	T	F	1-0.7=0.3
F	F	T	1-0.2=0.8
F	F	F	1-0.1=0.9

$f_4(J,A) = P(J|A)$

A	J	value
T	T	0.8
T	F	0.2
F	T	0.4
F	F	0.6

Step 2: Apply restrictions

$f_1(B) = P(B)$:

B	f_1 val
T	0.3
F	0.7

$f_2(E) = P(E)$

E	val
T	0.1
F	0.9

$f_3(A,B,E)$: $P(A|B,E)$

A	B	E	value
~~T~~	~~T~~	~~T~~	~~0.8~~
~~T~~	~~T~~	~~F~~	~~0.7~~
~~T~~	~~F~~	~~T~~	~~0.2~~
~~T~~	~~F~~	~~F~~	~~0.1~~
F	T	T	$1-0.8=0.2$
F	T	F	$1-0.7=0.3$
F	F	T	$1-0.2=0.8$
F	F	F	$1-0.1=0.9$

$f_4(J,A) = P(J|A)$

A	J	value
~~T~~	~~T~~	~~0.8~~
~~T~~	~~F~~	~~0.2~~
F	T	0.4
F	F	0.6

$f_5(B,E)$: $P(\sim a|E,B)$

B	E	value
T	T	0.2
T	F	0.3
F	T	0.8
F	F	0.9

$f_6(J)$: $P(J|A)$

J	value
T	0.4
F	0.6

Step 3: applying multiplicⁿ and summations

$$P(B, a=F) = f_1(B) \cdot \sum_E f_2(E) \cdot f_3(B,E) \cdot \underbrace{\sum_J f_6(J)}_{f_7()}$$

$\sum_J f_6(J)$: $f_6(J)$: $P(J|A)$

J	value
T	0.4
F	0.6

$\sum_J f_6(J) = 0.4 + 0.6 = 1$

$f_7()$

value
1

J gets eliminated

$$P(B, a=F) = f_1(B) \cdot f_7() \sum_E f_2(E) \cdot f_5(B,E)$$

$f_2(E) * f_5(B,E)$:

E	val
T	0.1
F	0.9

*

B	E	value
T	T	0.2
T	F	0.3
F	T	0.8
F	F	0.9

B	E	value
T	T	0.02
T	F	0.27
F	T	0.08
F	F	0.81

applying summation over E.

$f_8(B)$:

B	value
T	0.29
F	0.89

$$P(B, a=F) = f_1(B) \cdot f_7() \, f_8(B)$$

Step 4: multiply remaining factors

$$P(B, a=F) = f_1(B) \cdot f_7() \, f_8(B) \qquad f_{10}(B)$$

$f_1(B)$

B	f_1 val
T	0.3
F	0.7

$f_7()$

value
1

$f_8(B)$

B	value
T	0.29
F	0.89

=

B	value
T	0.087
F	0.623

$P(B, a=F) = f_{10}(B)$

Step 5: normalize

B	value
T	0.087
F	0.623

→

B	value
T	0.1225
F	0.8775

→ this is what is req.